

OPERATIONAL INTERVENTION MEMO
Last-Mile Delivery Defect Reduction
amazon_ecommerce_1M · 10 Lakh Orders · May 2026
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1. Context

This memo presents findings from a bottleneck audit of last-mile delivery operations conducted on the amazon_ecommerce_1M dataset — comprising 1,000,000 transactions across 5 Indian cities (Bangalore, Chennai, Delhi, Hyderabad, Mumbai). Analysis was performed using Power BI, segmenting delivery outcomes by geocoding risk tier, shipping load, and failure root cause. The objective was to identify leading predictors of delivery failure and recommend proactive interventions before failures occur.

2. Key Findings

Analysis of 10 lakh orders revealed the following:

- Overall Failure Rate: 12.20% across all cities and shipping bands.
- Top Root Cause: Time Window Missed accounts for 294,983 failures — 72% of all failure events.
- Geographic Concentration: Mumbai (10.74%) and Delhi (10.60%) show highest failure rates among High-Risk geocoding zones.
- Shipping Load Impact: Overloaded shipments fail at 20% vs 14% for Normal load — a 43% relative increase in failure risk.
- Seasonal Pattern: November records the highest monthly failure rate at 12.6% — consistent with festive season order surge and delivery overload.

3. Root Cause Analysis

Two compounding factors drive the majority of failures:

First, delivery time windows are too broad. When customers are given 4-hour windows with no real-time confirmation, absence rate increases significantly — directly causing Time Window Missed failures which account for 72% of all defects across 10 lakh orders.

Second, shipping overload degrades delivery quality. Overloaded shipments (shipping_time_days > 5) show a 43% higher failure rate than normal load shipments. This mirrors a pattern identified during Amazon operations (2019–2024) where team utilisation above 90% consistently correlated with quality score deterioration — from 98% down to 84% during peak periods.

Combined, these two factors create a failure cluster - high-volume periods with broad time windows = maximum defect concentration. November data confirms this directly.

4. Recommended Interventions

Based on root cause analysis, three interventions are recommended in order of priority:

Intervention 1 — Narrow Delivery Windows (High Priority) Reduce delivery windows from 4 hours to 2 hours for all orders in Mumbai and Delhi. Implement SMS confirmation 60 minutes before the attempt. Expected impact: 30-35% reduction in Time Window Missed failures.

Intervention 2 — Shipping Load Cap (High Priority) Cap daily order volume at 85% of maximum capacity during November and December. Redistribute excess volume to Hyderabad and Bangalore fulfilment centres which show lower failure rates. Expected impact: 20-25% reduction in overload-related failures.

Intervention 3 — Pre-Delivery Address Verification (Medium Priority) Flag all orders in High-Risk geocoding zones (Mumbai, Delhi) for address pre-verification before dispatch. This directly addresses Wrong Address or Return failures (115,990 orders) which account for 28% of all failures.

5. Business Impact

Current Cost of Failure:

At an estimated Rs.150 per re-delivery attempt, the current failure volume represents significant avoidable cost:

- Total Failed Orders: ~410,973 (12.20% of 1,000,000)
- Cost of Current Failures: Rs.150 x 410,973
= Rs. 6.16 Crore per cycle

Projected Savings from Interventions:

A conservative 25-30% failure reduction is projected based on a directly comparable intervention benchmark — Amazon last-mile geocoding accuracy improvement (2022-23) achieved 65% defect reduction through structured RCA and corrective action framework implementation. This project was led and executed by the analyst.

Applying a conservative 25% improvement:

- Failures reduced by: ~102,743 orders
- Cost saving: Rs.150 x 102,743 = Rs. 1.54 Crore per cycle
- Annually: Rs. 6.16 Crore in avoided redelivery cost

These projections are conservative. Full implementation of all three interventions is projected to achieve 30-35% failure reduction based on the Amazon benchmark, representing Rs. 7.2 Crore in annual savings.

6. Conclusion

This analysis demonstrates that last-mile delivery failures are not random — they are predictable and preventable. Time Window Missed and Shipping Overload are identifiable in advance using operational data. An early-warning system flagging high-risk zones and Overloaded periods enable proactive intervention before failures occur — shifting operations from reactive firefighting to data-driven prevention.

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